

Hugon Lee

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Research Statement

My research develops physics-grounded and data-efficient computational methods for predicting and designing the mechanical behavior of heterogeneous materials and their manufacturing processes. I integrate micromechanics, computational modeling, and machine learning to address complex material and process behavior under limited computational and experimental data, with current interests extending to physics-informed learning and AI-assisted scientific modeling.

Research Interests

Micromechanics & homogenization · Interface & interphase mechanics · Manufacturing process modeling · Data-efficient design & optimization · Physics-informed machine learning · Scientific AI

Appointments

Postdoctoral Fellow, InnoCORE PRISM-AI Center, KAIST

Aug. 2026 - Present

Faculty host: Prof. Seunghwa Ryu

Education

Ph.D. in Mechanical Engineering, KAIST

Feb. 2022 - Aug. 2026

Advisor: Prof. Seunghwa Ryu

Thesis: Micromechanical modeling and analysis of composites with interface and interphase effects

M.S. in Mechanical Engineering, KAIST

Feb. 2020 - Feb. 2022

Advisor: Prof. Seong Su Kim

Thesis: (A) study on permeability prediction of fiber-reinforced composite materials based on the representative volume element

B.S. in Mechanical Engineering, KAIST

Mar. 2014 - Feb. 2020

Military service: Sergeant, ROK Army/Korean Augmentation to The U.S. Army (KATUSA), Aug. 2016 - Apr. 2018.

Awards & Honors

Outstanding Paper Award, KSME

May 2026

Title: Multi-fidelity surrogates for efficient prediction of SFRP mechanical properties

Outstanding Paper Award, KSME CAE & Applied Mechanics Division

Apr 2025

Title: Modeling and simulation of DLP 3D printing process using Abaqus

Excellent Poster Presentation Award, KMSM

Dec 2022

Title: Efficient surrogate modeling of effective modulus for particle-reinforced composites with transfer learning framework

Academic Excellence Award, Mechanical Engineering, KAIST

Fall 2018 & Spring 2019

Dean's List, College of Engineering, KAIST

Fall 2018

In progress

18. Uncertainty-aware structure-property mapping of spinodoid metamaterials via heteroscedastic Gaussian process regression
Park, M.[†], Park, J.[†], Lee, M.[†], **Lee, H.**, Cho, H., Lee, I.^{*} and Ryu, S.^{*}
arXiv preprint (2026)
17. Machine learning-assisted predictive modelling of bubble growth phenomenon in foaming
Shen, X. W.[†], **Lee, H.**[†], Zhang, L., Chang, N. R.-S., Pehlert, G., Ryu, S.^{*}, Kweon, M. S. and Lee, P. C.^{*}
Submitted

Peer-reviewed

16. Toward depolyable AI surrogates for simulation-driven engineering
Park, D., **Lee, H.**, and Ryu, S.^{*}
Scientific Reports (2026)
15. Structural hierarchy can enhance the elastic modulus of random porous materials
Kim, Y., **Lee, H.**, Rhie, S., Ryu, S. and Kang, K.^{*}
Heliyon (2026)
14. Morphology-, noise-, and resolution-robust ultrasound elasticity imaging with Fourier neural operator
Kim, H., **Lee, H.**, Park, M. and Ryu, S.^{*}
Computer Methods and Programs in Biomedicine (2026)
13. Comparative assessment of composition- and structure-based surrogate models across 2D materials databases
Lee, I., Chae, H., Park, J., Shin, J., **Lee, H.**^{*} and Ryu, S.^{*}
Physical Chemistry Chemical Physics (2026)
12. Comparative study of Gaussian process and additive Gaussian process for multi-objective optimization in air-cooled battery thermal management systems
Cho, H., Kim, M., Park, Y., **Lee, H.**, Lee, I. and Ryu, S.^{*}
Journal of Mechanical Science and Technology (2025)
11. Toward knowledge-guided AI for inverse design in manufacturing: A perspective on domain, physics, and human-AI synergy (perspective article)
Lee, H.[†], Moon, H.[†], Lee, J.[†] and Ryu, S.^{*}
Advanced Intelligent Discovery (2025)
10. Sequential multi-objective Bayesian optimization of air-cooled battery thermal management system with spoiler integration
Cho, H., **Lee, H.** and Ryu, S.^{*}
Journal of Energy Storage (2025)
9. Bayesian optimization of tailgate rib structures enhancing structural stiffness under manufacturing constraints of injection molding
Lee, H.[†], Yeo, J.[†], Kong, K., Myeong, D., Jang, D., Lee, J., Choi, H., Kim, N.^{*} and Ryu, S.^{*}
Journal of Manufacturing Processes (2025)
8. Mean-field homogenization of liquid metal elastomer composites: comparative study and exact dilute solution of core-shell inclusion
Lee, H., Yeo, J., Yu, J., Moon, H. and Ryu, S.^{*}
Journal of Physics: Condensed Matter (2024)
7. Enhancing injection molding optimization for SFRPs through multi-fidelity data-driven approaches incorporating prior information in limited data environments
Lee, H.[†], Lee, M.[†], Jung, J., Lee, I.^{*} and Ryu, S.^{*}
Advanced Theory and Simulations (2024)

6. Data-driven prediction of strain fields in auxetic structures and non-contact validation with mechanoluminescence for structural health monitoring
Jo, J.[†], Park, M.[†], Kang, S.[†], **Lee, H.**[†], Gu, C.-Y., Kim, T.-S. and Ryu, S.*
International Journal of AI for Materials and Design (2024)
5. Comparative study of multi-objective Bayesian optimization and NSGA-III based approaches for injection molding process
Jung, J., Park, K., **Lee, H.**, Cho, B. and Ryu, S.*
Advanced Theory and Simulations (2024)
4. Innovative 3D printing of mechanoluminescent composites: vat photopolymerization meets machine learning
Jo, J.[†], Park, K.[†], Song, H.[†], **Lee, H.** and Ryu, S.*
Additive Manufacturing (2024)
3. Deep generative spatiotemporal learning for integrating fracture mechanics in composite materials: inverse design, discovery, and optimization
Park, D.[†], Lee, J.[†], **Lee, H.**, Gu, G. X. and Ryu, S.*
Materials Horizons (2024)
2. Advancements and challenges of micromechanics-based homogenization for the short fiber reinforced composites (review article)
Lee, H., Lee, S.* and Ryu, S.*
Multiscale Science and Engineering (2023)
1. Machine learning-based inverse design methods considering data characteristics and design space size in materials design and manufacturing: A review (review article)
Lee, J., Park, D., Lee, M., **Lee, H.**, Park, K., Lee, I. and Ryu, S.*
Materials Horizons (2023)

Selected Conference & Workshop Presentations

[†] Equal contribution *Corresponding author

- Multi-agent framework for micromechanical modeling, simulation, and applicability analysis
Lee, H. and Ryu, S.*
WCCM-ECCOMAS 2026 · Munich, Germany · Jul 2026 (Oral)
- Multi-fidelity surrogate for mechanical properties of short fiber reinforced plastics: leveraging information from theory to experiment
Lee, H. and Ryu, S.*
APCOM-ACCM 2025 · Brisbane, Australia · Dec 2025 (Oral)
- Micromechanical analysis on interphase effects of liquid metal elastomer composites
Lee, H. and Ryu, S.*
KSME 2024 · Jeju, Republic of Korea · Nov 2024 (Oral)
- Multi-fidelity predictive modeling of SFRP mechanical properties
Lee, H. and Ryu, S.*
KAIST-U of T International Doctoral Cluster (IDC) Workshop · Daejeon, Republic of Korea · Oct 2024
- Numerical implementation of digital light processing 3D printing process using Abaqus
Lee, H.[†], Jo, J.[†] and Ryu, S.*
ICTAM 2024 · Daegu, Republic of Korea · Aug 2024 (Oral)

Academic Talks

- **Invited group talk**, Prof. Patrick C. Lee group, University of Toronto, Canada
Introduction to multi-fidelity machine learning and its utilization for experimental data

Nov. 2023

Teaching & Mentoring

Teaching

- KAIST Industry-Academic Open Course, Republic of Korea Feb. 2026
Teaching assistant; led a one-hour lecture and hands-on tutorial on LLM usage and tool-calling for homogenization workflows
- 2026 Winter AI Class, Korean Society for Composite Materials (KSCM), Republic of Korea Feb. 2026
Teaching assistant; led a one-hour lecture and hands-on tutorial on LLM usage and tool-calling for homogenization workflows

Research Mentoring

- Mr. Inhyo Lee Jan. 2025 – Oct. 2025
Research design & manuscript preparation; published in *Physical Chemistry Chemical Physics*
- Mr. Sangwoo Jung Jun. 2024 – Oct. 2025
Master's research – modeling of DLP additive manufacturing process; manuscript in preparation
- Mr. Hanbin Cho Aug. 2023 – Aug. 2024
Master's research – numerical analysis & Bayesian optimization; published in *Journal of Energy Storage*
- Mr. Hyeonseong Jo Feb. 2021 – Feb. 2022
Master's research – experimental design, permeability analysis, numerical modeling

Professional Services

Journal Reviewer: Nature Communications, Archives of Computational Methods in Engineering, Mechanical Systems and Signal Processing, Applied Composite Materials, Progress in Additive Manufacturing, Journal of Thermal Analysis and Calorimetry, Discover Mechanical Engineering

Technical Skills

- **Programming:** Python, Fortran, MATLAB, $\LaTeX$
- **Simulation:** Abaqus, FEniCSx, COMSOL, HyperWorks, Moldflow, ANSYS Fluent
- **ML & AI:** PyTorch, GPy, LangChain, LangGraph
- **Experimental Design & Validation:** Material and process characterization, experimental protocol design, simulation-experiment comparison
- **Visualization & Workflow:** Illustrator, ParaView, PyVista, Git
- **Languages:** Korean (native), English (fluent)

References

Dr. Seunghwa Ryu (Ph.D. advisor)
Head, Dept. of AX, KAIST
Professor, Dept. of Mechanical Engineering, KAIST

Dr. Patrick C. Lee
Associate Professor, Dept. of Mechanical and Industrial Engineering, University of Toronto

Dr. Sangryun Lee
Assistant Professor, Dept. of Mechanical and Biomedical Engineering, Ewha Womans University